Fiscal Year:	FY 2018	**Task Last Updated:**	FY 08/22/2017
PI Name:	Bailey, Susan M. Ph.D.		
Project Title:	Assessing Telomere Lengths and Telomerase Activity in Astronauts		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Fort Collins	**State:**	CO
Zip Code:	80523-1618	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Crew Health NNJ12ZSA002N
Start Date:	10/24/2013	**End Date:**	12/23/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date is 12/23/2018 per NSSC information (Ed., 6/20/18) NOTE: End date is 6/23/2018 per NSSC information and PI (Ed., 8/2/17) NOTE: End date changed to 10/23/2017 per NSSC information (Ed., 7/26/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX14AB02G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The ends of human chromosomes are capped by telomeres, tandem arrays of repetitive DNA sequence and associated proteins, which serve to protect chromosomal termini from inappropriate degradation. Functional telomeres also prevent these natural DNA ends from being recognized as broken DNA (double-strand breaks; DSBs) and triggering inappropriate DNA damage responses, which could result in chromosomal instability. While it is well established that telomere length erodes with normal aging, it is becoming increasingly appreciated that telomere length is also influenced by a variety of other factors including life stress, infection, and inflammation. Inflammation also results in oxidative stress, both being known contributors to telomere erosion. Recent research supports telomere maintenance as a key integrating component for the cumulative effects of genetic, environmental, and lifestyle factors on aging and aging-related diseases. The rate of telomere length change, therefore, provides an informative biomarker of biological aging. Further, dysfunction/decline of telomere length can be linked to age-related pathologies, ranging from reduced immune function and loss of fertility, to cardiovascular disease and cancer. Preservation of telomere length requires telomerase, the specialized reverse transcriptase capable of de novo addition of telomeric repeats onto the very ends of newly synthesized telomeres. Telomerase activity is prominent in highly proliferative cells like stem, germline, and cancer, but virtually absent in normal somatic cells. Changes in diet and lifestyle have been shown to modulate telomerase activity, with healthy lifestyles and diets being positively correlated with telomere length. We've demonstrated that ionizing radiation also influences telomeres and telomerase. However, the relationships between telomere length and telomerase activity (measured in blood lymphocytes), lifestyle factors, and environmental exposures are largely unknown. We speculate that for the astronauts, telomere maintenance represents a particularly relevant biomarker, as it reflects the combined exposures and experiences encountered during space travel. That is, an individual astronaut's genetic susceptibilities, exposures to galactic space radiations, nutritional, physical, and psychological stressors encountered, are all captured as changes of telomere length maintenance; telomeres don't forget! Our overall goal is to identify and begin to define the risk of accelerated telomere shortening and changes in telomerase activity associated with spaceflight, a lifestyle with a variety of unique stressors. Recruitment of prospective astronauts (and age, gender-matched ground control subjects) is complete, and monitoring of crewmember's telomere length and telomerase activity is on-going; samples are being collected pre-flight (to establish baseline) and post-flight (to evaluate long-term/permanent changes). Together with data sharing related to e.g., diet and nutrition, oxidative stress and radiation exposure, results will help to establish relevant relationships and suggest potential interventions for further study.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although relationships between telomere length and telomerase activity, and lifestyle factors (e.g., nutritional, physical, and psychological stressors) underlie aging and age-related degenerative pathologies (e.g., cardiovascular disease and cancer), they are largely unknown for human populations. Certainly aging and such age-related pathologies are an everyday concern on Earth as well as in space. The rate at which telomeres shorten provides an informative biomarker of biological aging that is influenced by a variety of lifestyle factors, including a variety of stressors and environmental exposures (e.g., UV and ionizing radiations). These on-going studies seek to make comparisons with and between unrelated astronauts (and age/gender-matched controls) that will serve to identify spaceflight and individual susceptibility factors that affect telomere length and telomerase activity. Results will begin to establish relevant relationships and suggest potential mitigation strategies for future study; therefore, they have the potential to improve overall health not only of astronauts, but of those on Earth as well.
Task Progress:	We continued with post-flight collections of astronaut and age/gender-matched ground control subject samples; all were processed and shipped successfully. Sample collections have been completed for 5 astronauts. One year debriefs were conducted for two Telomeres astronauts. Telomere length and telomerase activity analyses are on-going, being performed individually and in batch. Chromosome analyses are also still in progress and being finalized as R+270 samples are received. Despite our continued efforts and discussions, we were still unable to obtain permission for analyzing any retrospective samples.
Bibliography Type:	Description: (Last Updated: 12/10/2025)
Articles in Peer-reviewed Journals	Schmidt MA, Bailey SM, Goodwin TJ, Jones JA, Killian JP, Legato MJ, Limoli C, Moussa S, Ploutz-Snyder L. "Men, Women, and Space Travel: gene-linked molecular networks, human countermeasures, and legal and ethical considerations." Gender and the Genome. 2017 Jun;1(2):54-67. Roundtable discussion. https://doi.org/10.1089/gg.2017.29004.rtl , Jun-2017
Articles in Peer-reviewed Journals	McKenna MJ, Bailey SM. "Chromosomal and telomeric biomarkers of normal tissue injury to evaluate risk of degenerative health effects (secondary malignancy, cardiovascular disease) post radiation therapy." Transl Cancer Res 2017 Jul;6(Suppl 5):S789-S794. https://doi.org/10.21037/tcr.2017.05.20 , Jul-2017
Articles in Peer-reviewed Journals	Bezdan D, Grigorev K, Meydan C, Pelissier Vatter FA, Cioffi M, Rao V, MacKay M, Nakahira K, Burnham P, Afshinnekoo E, Westover C, Butler D, Moszary C, Donahoe T, Foox J, Mishra T, Lucotti S, Rana BK, Melnick AM, Zhang H, Matei I, Kelsen D, Yu K, Lyden DC, Taylor L, Bailey SM, Snyder MP, Garrett-Bakelman FE, Ossowski S, De Vlaminck I, Mason CE. "Cell-free DNA (cfDNA) and exosome profiling from a year-long human spaceflight reveals circulating biomarkers." iScience. 2020 Dec 18;23(12):101844. https://doi.org/10.1016/j.isci.2020.101844 ; PMID: 33376973; PMCID: PMC7756145 , Dec-2020